Krishnaa **Vadivel** | Software Engineer | Machine Learning | Distributed Computing

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Profile

Software engineer and PhD researcher focused on machine learning, distributed computing, numerical software, and engineering tooling. Strong background in Python and C++ development for simulation, automation, GPU acceleration, data systems, and infrastructure spanning research and production-adjacent environments.

Research Experience

Yale University, Electrical Engineering
Research Assistant / PhD Researcher *2021–present*

- Develop scientific software for first-principles and many-body workflows in materials research, combining Python and C++ with MPI, OpenMP, PETSc, SLEPc, CUDA, ROCm, and Linux-based automation.
- Build distributed and accelerator-enabled workflows for HPC environments including Frontier, Frontera, and Perlmutter, with experience spanning NVIDIA and AMD GPU systems on national-lab clusters.
- Create reusable tooling, notes, and structured workflows that bridge theory, implementation, debugging, and reproducibility in computational science.
- Extend into ML-assisted and retrieval-driven automation where useful, particularly with LangChain, OpenAI API models, and local Ollama models for coding and research-workflow setup.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on self-trapped exciton formation and related first-principles work.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Built CUDA-accelerated processing software for large acoustic waveform and sensor datasets used in engineering diagnostics and field-tool analysis.
- Developed ASP.NET services and Microsoft SQL Server workflows for internal software, field-data management, and remote tool-calibration support.
- Developed C# GUI applications for preparing customer inputs and interacting with engineering workflows tied to deployed hardware systems.
- Worked close to embedded and instrumentation systems, giving practical experience with software operating alongside deployed hardware and field constraints.

FindLight
Photonics Technical Writer *2018–2019*

- Produced technical content for optics and photonics topics, demonstrating strong technical communication across specialized engineering subject matter.

Selected Projects

Equivariant Graph Neural Networks for Molecular Modeling

- Built PyTorch Geometric workflows for molecular-property prediction using equivariant graph neural networks and graph-transformer-style modeling ideas on QM9-style data.

Agentic Coding and Scientific Workflow Generation

- Developed LangChain-based retrieval-driven tooling using OpenAI API models and local Ollama models to read documentation and academic papers, then generate structured starter inputs and workflow artifacts.

Scientific Visualization and GPU Tooling

- Built CUDA-based tooling and related utilities for large numerical datasets, interactive inspection, and engineering communication.

Photonics and Simulation Code Artifacts

- Created clean, inspectable software artifacts for FDTD-based photonics simulation and related numerical-science workflows.

Education

Yale University
PhD, Electrical Engineering *2021–present*
Ongoing work combining scientific computing, high-performance workflows, and computational materials research.

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Selected Coursework

Computing Foundations: Theoretical Computer Science; Probabilistic Machine Learning

Scientific Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Engineering Foundations: Signals and systems, embedded systems, semiconductor physics, and numerical/scientific computing practice

Technical Skills

Languages: Python, C, C++, JavaScript, Bash, PowerShell, SQL

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, NumPy, pandas, SciPy, SymPy, matplotlib

Systems: Linux command line, CMake, Docker, Docker Compose, Kubernetes, gcloud

Platforms: CUDA, ROCm, Frontier, Frontera, Perlmutter

ML / Automation: PyTorch, PyTorch Geometric, scikit-learn, LangChain, OpenAI API, Ollama, graph neural networks, graph transformers, agentic workflows